

# **AI-POWERED DRONE SURVEILLANCE SYSTEM FOR STUDENT ACTIVITY MONITORING**

A thesis submitted in partial fulfillment of the  
requirement for the award of the degree of

## **Bachelor of Technology in Electronics & Communication Engineering**

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**2026**

## Department of Electronics & Communication Engineering



### CERTIFICATE

This is to certify that the thesis report entitled “**AI-POWERED DRONE SURVEILLANCE SYSTEM FOR STUDENT ACTIVITY MONITORING**” is being submitted by “**D. Vamsi (Y22EC034), B. Praneeth koundinya (Y22EC021), G. Harika (Y22EC047), M. Mahendra (Y21EC102)**” in partial fulfillment for the award of the Degree of **Bachelor of Technology in Electronics & Communication Engineering** to R.V.R & J.C College of Engineering (Autonomous) is a record of bonafide work carried out by them under my guidance and supervision. The results embodied in this project report have not been submitted to any other University or Institute for the award of any degree or diploma.

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# Abstract

Maintaining safety and discipline within campus environments has become increasingly challenging due to the limitations of conventional surveillance systems, which are often restricted by fixed coverage and require continuous human monitoring. In this context, the integration of unmanned aerial vehicles (UAVs) with intelligent data processing offers a promising solution for dynamic and automated surveillance. This project presents the design and implementation of an intelligent drone-based monitoring system capable of real-time video acquisition, analysis, and alert generation. The system utilizes a manually controlled drone equipped with an ESP32-S3 module, which captures and streams live video over a local network established using a mobile hotspot. This approach ensures a lightweight and cost-effective hardware configuration without the need for onboard high-performance processing units.

The streamed video is transmitted to a cloud-based environment, specifically Google Colab, where machine learning algorithms are executed. The system is structured into two primary analytical modules: face recognition and action recognition. The face recognition module identifies individuals present in the surveillance area, while the action recognition module analyzes human behavior to detect indisciplinary or suspicious activities. Upon detection of such activities, the system generates alerts to notify the concerned authorities, enabling timely intervention. By offloading computational tasks to the cloud, the proposed system achieves efficient real-time processing while maintaining scalability and flexibility. The overall system demonstrates the potential of combining drone technology with machine learning to create an intelligent, responsive, and cost-efficient surveillance solution.

# Acknowledgements

We would like to express our sincere gratitude to the **Department of Electronics & Communication Engineering**, R.V.R. & J.C. College of Engineering, for providing the necessary facilities, infrastructure, and a supportive environment to successfully carry out this project. We are thankful to **Dr. Kolla Srinivas**, Principal, R.V.R. & J.C. College of Engineering, for his constant encouragement and motivation. We also extend our heartfelt thanks to **Dr. T. Ranga Babu**, Head of the Department, for his valuable guidance, suggestions, and continuous support throughout the development of this work.

We are especially grateful to **Dr. M. Himaja**, Guide, for her valuable mentorship and guidance throughout the practical aspects of the project. We would also like to express our sincere appreciation to our families, friends, and well-wishers for their constant support, encouragement, and understanding, which played a vital role in the successful completion of this project.

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# 1

## INTRODUCTION

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## 1.1 Overview

The increasing demand for safety and discipline in educational institutions has led to the development of advanced surveillance systems. Traditional monitoring methods, such as fixed CCTV cameras, are limited in coverage and require continuous human supervision. These limitations make it difficult to effectively monitor large and dynamic environments such as college campuses.

To overcome these challenges, this project proposes an drone-based monitoring system that combines the mobility of unmanned aerial vehicles with the analytical capabilities of machine learning. The system utilizes a manually controlled drone equipped with an ESP32-S3 module for real-time video capture. The captured video is transmitted over a local network using a mobile hotspot and is processed on a local computing system, enabling efficient real-time analysis.

The system is designed with two primary modules: face recognition and action recognition. The face recognition module identifies individuals present in the monitored area using the DeepFace framework, which extracts facial embeddings for accurate identification. The action recognition module analyzes human behavior by processing consecutive video frames and classifies activities using a lightweight multi-class classification approach.

Upon detecting any indisciplinary or suspicious activity, the system generates alerts to notify the concerned authorities. The proposed approach emphasizes real-time performance, computational efficiency, and practical deployability, making it suitable for intelligent campus surveillance applications.

## 1.2 Problem statement

Traditional surveillance systems, such as CCTV cameras, are limited by their fixed positions and inability to dynamically monitor large areas. These systems require continuous human supervision, making them inefficient and prone to delayed response in identifying indisciplinary activities.

In campus environments, maintaining discipline and ensuring safety is challenging due to the wide coverage area and lack of intelligent monitoring mechanisms. Existing systems do not provide automated analysis of activities or real-time alerts, which reduces their overall effectiveness.

Therefore, there is a need for a system that can provide flexible monitoring, perform real-time analysis of human behavior, and automatically detect and report indisciplinary activities with minimal human intervention.

## 1.3 Objectives

The main objectives of this project are:

- To design and develop a drone-based monitoring system for campus surveillance.
- To implement real-time video capture using an ESP32-S3 module.
- To perform face recognition for identifying individuals using a pretrained model.
- To implement action recognition for detecting indisciplinary activities.
- To process video data efficiently using a lightweight machine learning approach.
- To generate alerts for notifying authorities in case of suspicious behavior.

## 1.4 Scope of the project

The scope of this project is to design and develop a drone-based monitoring system capable of real-time surveillance and automated activity analysis within a campus environment. The system utilizes a manually controlled quadcopter platform equipped with an ESP32-S3 module to capture live video, which is transmitted over a local wireless network using a mobile hotspot.

The captured video is processed on a local computing system, where machine learning models are employed for face recognition and action recognition. The system is designed to identify individuals and analyze their activities to detect indisciplinary behavior. Upon detection, alerts are generated to notify the concerned authorities.

The implementation is constrained to manual drone operation and offboard processing, thereby avoiding the need for high-performance onboard computation. The system focuses on real-time performance, computational efficiency, and ease of deployment within a campus environment.

The scope of the project does not include autonomous drone navigation, advanced multi-object tracking, or fully integrated real-time alert delivery systems. However, these features are considered as potential extensions for future work to enhance the system's capabilities and scalability.

## 1.5 Organization of the report

This report is organized into several chapters, each describing different aspects of the proposed system.

- Chapter 1 presents the introduction, including overview, problem statement, objectives, and scope of the project.
- Chapter 2 discusses the literature survey and reviews existing methods related to drone-based monitoring and activity recognition.
- Chapter 3 describes the system architecture and overall design of the proposed system.
- Chapter 4 explains the methodology and working process of the system.
- Chapter 5 details the implementation of both hardware and software components.
- Chapter 6 presents the results obtained and discusses the performance of the system.
- Chapter 7 concludes the work and suggests possible future enhancements.

# 2

## LITERATURE SURVEY

---

This chapter presents a comprehensive review of existing research related to intelligent surveillance systems, including drone-based monitoring, face recognition, and human action recognition. Recent advancements in artificial intelligence have enabled the development of automated systems capable of analyzing visual data and assisting in real-time decision-making [1].

Modern surveillance systems aim to improve safety and monitoring efficiency by reducing human dependency and enabling automated detection of suspicious activities. However, several challenges such as computational complexity, real-time processing, and integration of multiple modules still exist [2]. These challenges motivate the need for efficient and scalable solutions.

## **2.1 Drone-based surveillance systems**

**2.1.1 Overview of drone surveillance** Drone-based surveillance systems have emerged as an effective alternative to traditional fixed-camera systems due to their ability to provide dynamic coverage and mobility. These systems can monitor large and complex environments such as campuses, public spaces, and disaster zones [3].

Unlike stationary cameras, drones can access difficult-to-reach areas and provide aerial perspectives, which significantly enhances situational awareness. This flexibility makes drone-based systems highly suitable for modern surveillance applications.

**2.1.2 Applications of drone surveillance** Drones are widely used in applications such as crowd monitoring, traffic control, border surveillance, and emergency response systems [4]. These systems collect real-time visual data, which can be analyzed either manually or through automated algorithms.

However, many existing implementations still rely on human monitoring, which introduces delays and reduces overall efficiency. To address this limitation, intelligent systems capable of automated analysis are required [5].

## **2.2 Face recognition techniques**

**2.2.1 Traditional methods** Traditional face recognition techniques such as Eigenfaces and Fisherfaces are based on statistical feature extraction methods. These approaches transform facial images into lower-dimensional feature spaces and identify patterns using linear models [6].

Although computationally efficient, these methods are highly sensitive to variations in

illumination, pose, and facial expressions. As a result, their performance is limited in real-world scenarios where such variations are unavoidable.

**2.2.2 Deep learning approaches** Recent advancements in deep learning have significantly improved the accuracy and robustness of face recognition systems. Convolutional Neural Networks (CNNs) are capable of learning complex feature representations directly from images, enabling reliable identification under varying conditions [7].

Modern face recognition systems often use embedding-based approaches, where facial features are represented as numerical vectors. These embeddings are compared using similarity measures, allowing efficient and scalable identification [8]. Such approaches have become the standard in real-time recognition systems.

## **2.3 Action recognition techniques**

**2.3.1 Conventional methods** Early action recognition techniques relied on handcrafted features and motion-based analysis methods. These approaches focused on detecting variations in pixel intensities and movement between consecutive frames [9].

While computationally simple, these methods are limited in their ability to capture complex temporal relationships and higher-level motion patterns, making them less effective for real-world applications.

**2.3.2 Deep learning-based methods** Deep learning-based approaches have significantly advanced the field of action recognition by enabling the analysis of both spatial and temporal information. Models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are commonly used to extract features and model sequential data [10].

These models can recognize complex human activities with high accuracy. However, they require large datasets, high computational power, and specialized hardware, which limits their applicability in real-time and resource-constrained environments.

**2.3.3 Lightweight approaches** To overcome the limitations of computationally intensive models, lightweight action recognition techniques have been developed. These approaches focus on extracting simple motion-based features and using efficient classification algorithms to achieve faster processing speeds.

Although lightweight methods may not achieve the same level of accuracy as deep learning models, they provide a practical balance between performance and computational effi-

ciency, making them suitable for real-time surveillance systems.

## 2.4 Limitations of existing systems

Despite significant advancements, existing surveillance systems still face several limitations. Traditional systems based on fixed cameras lack flexibility and cannot provide complete coverage of dynamic environments. Drone-based systems improve mobility but often lack real-time intelligent analysis capabilities [3].

Additionally, advanced deep learning-based solutions require high computational resources, making them costly and difficult to deploy in practical scenarios [2]. Another major limitation is the lack of integration between face recognition and action recognition within a single efficient framework.

These challenges highlight the need for an integrated surveillance system that combines mobility, real-time processing, and computational efficiency.

## 2.5 Summary

This chapter presented a detailed review of existing approaches in drone-based surveillance, face recognition, and action recognition. The analysis highlighted the strengths and limitations of current systems and emphasized the need for efficient and scalable solutions.

The proposed system addresses these challenges by integrating drone-based monitoring with lightweight machine learning techniques to achieve real-time face recognition and action detection, ensuring improved performance and practical deployment.

## 2.6 Comparison of Existing Work and Proposed System

The comparison focuses on analyzing different surveillance approaches based on key factors such as methodology, computational efficiency, accuracy, and real-time applicability. Existing systems in the literature employ a wide range of techniques including IoT-based monitoring, deep learning models, and drone-assisted surveillance.

Several studies emphasize high accuracy by using advanced deep learning architectures such as CNN and LSTM models. While these approaches are effective in recognizing complex patterns and activities, they often require high computational resources, large datasets, and specialized hardware, which limits their practical deployment in real-time environments.

On the other hand, some systems adopt lightweight techniques such as motion-based analysis and traditional image processing methods. These approaches offer faster execution and lower computational requirements, but they may lack the ability to accurately detect complex activities or perform robust recognition under varying conditions.

The following table presents a detailed comparison highlighting the differences between existing approaches and the proposed system in terms of their advantages and limitations.

Table 2.1: Comparison of Existing Work and Proposed System

| System                 | Approach                                                                 | Advantages                                                     | Limitations                                        |
|------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------|
| Existing Work [1]      | AI-based surveillance system                                             | Enables automated monitoring and analysis                      | Requires high computational resources              |
| Existing Work [2]      | IoT-based surveillance system                                            | Supports remote monitoring and connectivity                    | Limited processing capability                      |
| Existing Work [3]      | Motion-based action recognition                                          | Fast and simple implementation                                 | Limited accuracy for complex activities            |
| Existing Work [4]      | Traditional face recognition                                             | Computationally efficient                                      | Sensitive to lighting and pose variations          |
| Existing Work [5]      | CNN-LSTM based recognition                                               | High accuracy in activity detection                            | High computational complexity                      |
| Existing Work [6]      | Smart surveillance system                                                | Scalable and wide coverage                                     | Expensive infrastructure                           |
| Existing Work [7]      | Drone-based surveillance                                                 | Flexible and mobile monitoring                                 | Requires manual control                            |
| Existing Work [8]      | Deep learning-based systems                                              | Accurate detection and classification                          | Requires large datasets and hardware               |
| Existing Work [9]      | DeepFace/Dlib recognition                                                | High robustness and accuracy                                   | Computational overhead                             |
| Existing Work [10]     | LSTM-based sequence models                                               | Captures temporal dependencies                                 | Longer training time                               |
| <b>Proposed System</b> | Drone-based monitoring with DeepFace and motion-based action recognition | Real-time detection, low computational cost, integrated system | Moderate accuracy compared to deep learning models |

# 3

## SYSTEM ARCHITECTURE

---

### 3.1 Introduction

This chapter presents the overall architecture of the proposed drone-based monitoring system. It describes how various hardware and software components interact to perform real-time video capture, processing, and activity detection.

The architecture illustrates the flow of data from the drone-mounted camera to the processing unit, where face recognition and action recognition modules operate to analyze the captured video and generate alerts.

### 3.2 Overall system design

The proposed system is designed as an integrated drone-based surveillance framework that enables real-time monitoring and intelligent analysis of activities within the campus environment. The system combines hardware components for data acquisition with software modules for processing and decision-making.

The system consists of a manually controlled drone equipped with an ESP32-S3 module for real-time video capture. The drone is used to monitor the campus area from different viewpoints, providing flexible and dynamic coverage. The ESP32-S3 module captures the video stream and transmits it over a wireless network using a mobile hotspot, ensuring continuous communication with the processing unit.

The transmitted video stream is received by a local computing system, where it is processed frame by frame. Each frame is analyzed using machine learning techniques to extract relevant information for further decision-making. The processing pipeline is designed to operate efficiently in real time, enabling immediate analysis of the monitored environment.

The system employs two parallel modules: face recognition and action recognition. The face recognition module utilizes the DeepFace framework to identify individuals by comparing facial embeddings with a pre-stored database. This allows the system to recognize known individuals as well as detect unknown persons.

Simultaneously, the action recognition module analyzes human behavior by extracting motion-based features from consecutive frames. By measuring frame differences and applying threshold-based classification, the system categorizes activities into normal, aggressive, and fight scenarios. This module is optimized for real-time performance with minimal computational overhead.

Both modules operate concurrently and their outputs are combined to provide context-aware analysis. By associating detected actions with identified individuals, the system en-

hances the accuracy and relevance of surveillance. For example, identifying a specific individual involved in aggressive behavior improves the effectiveness of monitoring. When any indisciplinary or suspicious activity is detected, the system generates an alert in real time. Overall, the system design focuses on achieving a balance between accuracy, computational efficiency, and practical deployment. The integration of drone mobility, real-time video streaming, and intelligent analysis makes the proposed system suitable for modern surveillance applications.

The overall architecture of the system is shown in Figure 3.1.

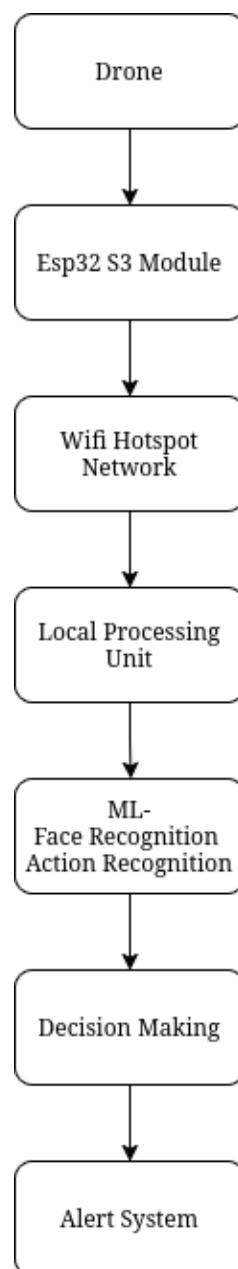


Figure 3.1: System architecture of the proposed system

# 4

## **METHODOLOGY**

---

## 4.1 Introduction

The proposed system integrates both hardware and software components to achieve efficient and reliable monitoring. A manually controlled drone equipped with an ESP32-S3 module is used for capturing live video data. This data is transmitted wirelessly through a mobile hotspot to a local computing system, where further processing is carried out. The continuous video stream is divided into frames, which serve as input for the machine learning modules.

The core functionality of the system is based on two parallel processing modules: face recognition and action recognition. The face recognition module identifies individuals by extracting facial features using a pretrained DeepFace model and comparing them with a stored database. Simultaneously, the action recognition module analyzes motion patterns across consecutive frames to classify human activities.

The integration of these modules enables the system to perform context-aware analysis by associating detected activities with identified individuals. Based on this combined analysis, the system determines whether the observed behavior is normal or indisciplinary. In cases of suspicious or aggressive activity, an alert is generated to notify the concerned authorities.

Furthermore, the methodology emphasizes a balance between computational efficiency and detection accuracy. Instead of relying on computationally expensive models, the system adopts optimized techniques that support real-time processing while maintaining satisfactory performance.

Overall, this methodology provides a practical and scalable framework for implementing intelligent drone-based surveillance systems in real-world environments such as educational institutions.

## 4.2 System workflow

The proposed system follows a structured workflow based on real-time video acquisition and processing. The drone is manually controlled to monitor the target area, and the ESP32-S3 module captures live video, which is transmitted over a wireless network using a mobile hotspot. The video stream is received and processed on a local computing system. The incoming video is divided into consecutive frames, which are analyzed using machine learning techniques.

The system consists of two parallel processing modules: face recognition and action recognition. The face recognition module detects faces from each frame and identifies

individuals using a pretrained DeepFace model. The action recognition module analyzes motion patterns across consecutive frames by extracting motion-based features such as intensity and variation. Initially calibration phase is used to determine baseline motion levels in the environment. Based on this, adaptive thresholds are applied to classify activities into three categories: *normal*, *aggressive*, and *fight*. To improve stability, temporal smoothing is applied using recent predictions to reduce fluctuations in classification.

Based on the outputs of both modules, the system determines whether the observed behavior is normal or indisciplinary. This workflow ensures efficient real-time processing while maintaining a balance between computational complexity and detection accuracy. The workflow of the proposed system is illustrated in Figure 4.1.

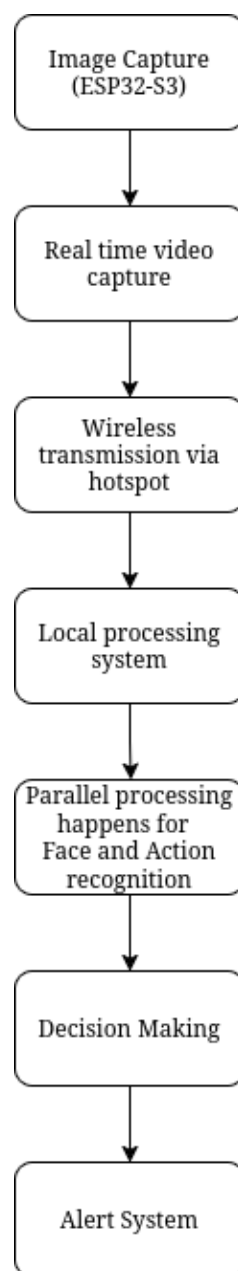


Figure 4.1: Workflow of the proposed system

### 4.3 Image capture and transmission

The ESP32-S3 module captures real-time video from the drone-mounted camera. The captured video is transmitted over a local wireless network using a mobile hotspot, enabling continuous data transfer to the processing system.

The video stream is received by a local computing system, where it is processed in real time. The incoming video is divided into consecutive frames for analysis by the machine learning modules.

This approach ensures efficient data transmission while supporting real-time processing and analysis of the monitored environment.

### 4.4 Face recognition process

The face recognition module is used to identify individuals present in the video stream. Each frame obtained from the video is analyzed to detect faces using a face detection algorithm.

For each detected face, feature extraction is performed using a pretrained DeepFace model (VGG-Face), which generates a numerical representation known as an embedding. These embeddings are normalized to ensure consistency and robustness.

A database of known individuals is maintained by storing average embeddings computed from multiple images per person. During recognition, the embedding of the detected face is compared with the stored embeddings using distance-based matching.

The identity corresponding to the minimum distance is selected, and a threshold is applied to determine whether the face belongs to a known individual or is classified as *Unknown*. The recognized identity is then displayed on the video output along with a bounding box.

This module enables accurate identification of individuals and helps in associating detected activities with specific persons.

The similarity between two face embeddings is computed using Euclidean distance:

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (4.1)$$

where  $x_i$  and  $y_i$  represent the embedding vectors of two faces, and  $d$  denotes the Euclidean distance between them. A threshold is applied to determine whether the faces belong to the same individual.

## 4.5 Action recognition process

The action recognition module analyzes human activities by processing consecutive video frames. Instead of relying on individual images, the system captures temporal information by examining motion patterns across multiple frames.

The incoming video frames are converted to grayscale and stored in a short buffer. Motion is computed using frame differencing techniques, where the absolute difference between consecutive frames is used to estimate movement intensity.

An initial calibration phase is performed to determine baseline motion levels in the environment. Based on this calibration, adaptive thresholds are defined to classify activities into three categories: *normal*, *aggressive*, and *fight*.

To improve prediction stability, temporal smoothing is applied using a sliding window of recent predictions. The final action label is determined using majority voting, which reduces noise and false detections.

This approach enables efficient real-time action recognition while maintaining a balance between computational complexity and detection accuracy.

Motion between consecutive frames is calculated using frame differencing:

$$M = \sum |F_t - F_{t-1}| \quad (4.2)$$

where  $F_t$  and  $F_{t-1}$  represent consecutive grayscale frames. The computed motion value is used to classify activities.

$$\text{Activity} = \begin{cases} \text{Normal} & M < T_1 \\ \text{Aggressive} & T_1 \leq M < T_2 \\ \text{Fight} & M \geq T_2 \end{cases} \quad (4.3)$$

## 4.6 Alert generation

When the system detects an indisciplinary activity, particularly in cases classified as *aggressive* or *fight*, it generates an alert to notify the concerned authorities. This enables timely intervention and improves overall surveillance effectiveness.

The alert is triggered based on the output of the action recognition module and is further supported by the identification results from the face recognition module. The system can provide information such as the detected activity and the identity of the individual involved.

# 5

## IMPLEMENTATION

---

## 5.1 Introduction

This chapter presents the implementation of the proposed drone-based monitoring system. The system integrates real-time video acquisition with face recognition and action recognition to perform intelligent surveillance. The implementation focuses on processing live video data, identifying individuals using DeepFace, and classifying human activities using motion-based analysis.

The system is designed to operate efficiently in real-time by combining hardware components with optimized software modules, enabling continuous monitoring and automated detection of indisciplinary activities.

## 5.2 Face recognition implementation

The face recognition module is implemented using the DeepFace framework with the VGG-Face model, which generates high-dimensional facial embeddings for identification. A database of known individuals is created by extracting embeddings from multiple images per person and computing a representative mean embedding.

During runtime, each incoming video frame is analyzed to detect faces. For every detected face, an embedding vector is generated and normalized. This embedding is then compared with the stored database using Euclidean distance. If the distance between embeddings falls below a predefined threshold, the individual is recognized; otherwise, the face is labeled as *Unknown*. The identified name is displayed along with a bounding box on the video frame.

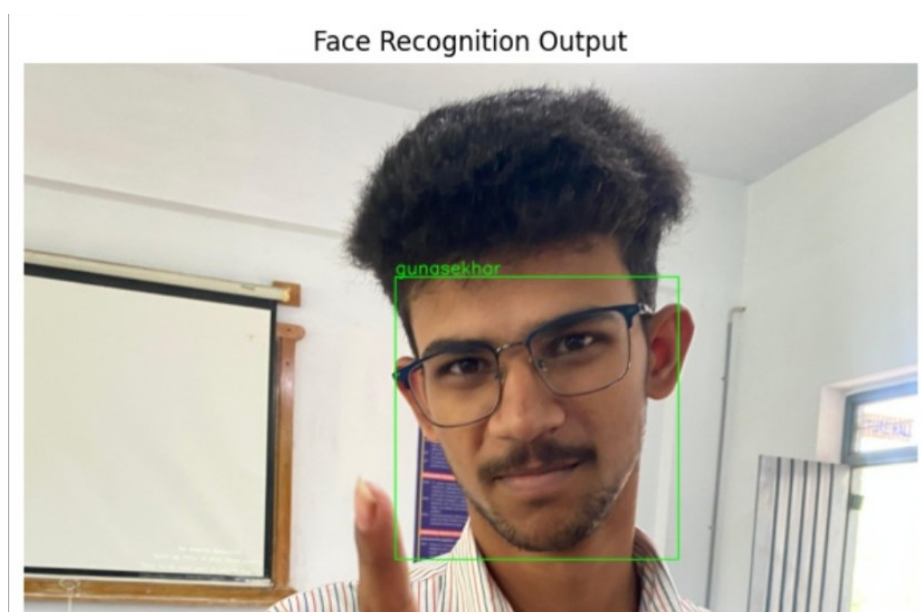


Figure 5.1: Face recognition output



Figure 5.2: Face recognition output showing multiple individuals detected and identified

This approach enables accurate real-time identification of multiple individuals in a scene while maintaining robustness under moderate variations in lighting and facial orientation.

### 5.3 Action recognition implementation

The action recognition module is designed to classify human activities based on motion patterns observed in consecutive video frames. The system computes motion intensity using frame differencing between consecutive grayscale frames.

An initial calibration phase is performed to determine baseline motion levels in the environment. Based on this, adaptive thresholds are defined to classify activities into three categories: *normal*, *aggressive*, and *fight*. To enhance prediction stability, temporal smoothing is applied using a sliding window mechanism, where recent predictions are stored and the final output is determined through majority voting.

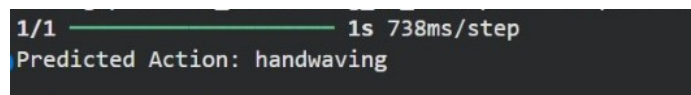


Figure 5.3: Action recognition output showing hand waving detection

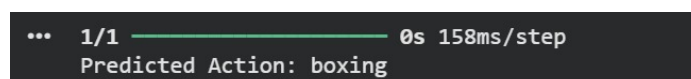


Figure 5.4: Action recognition output showing boxing detection

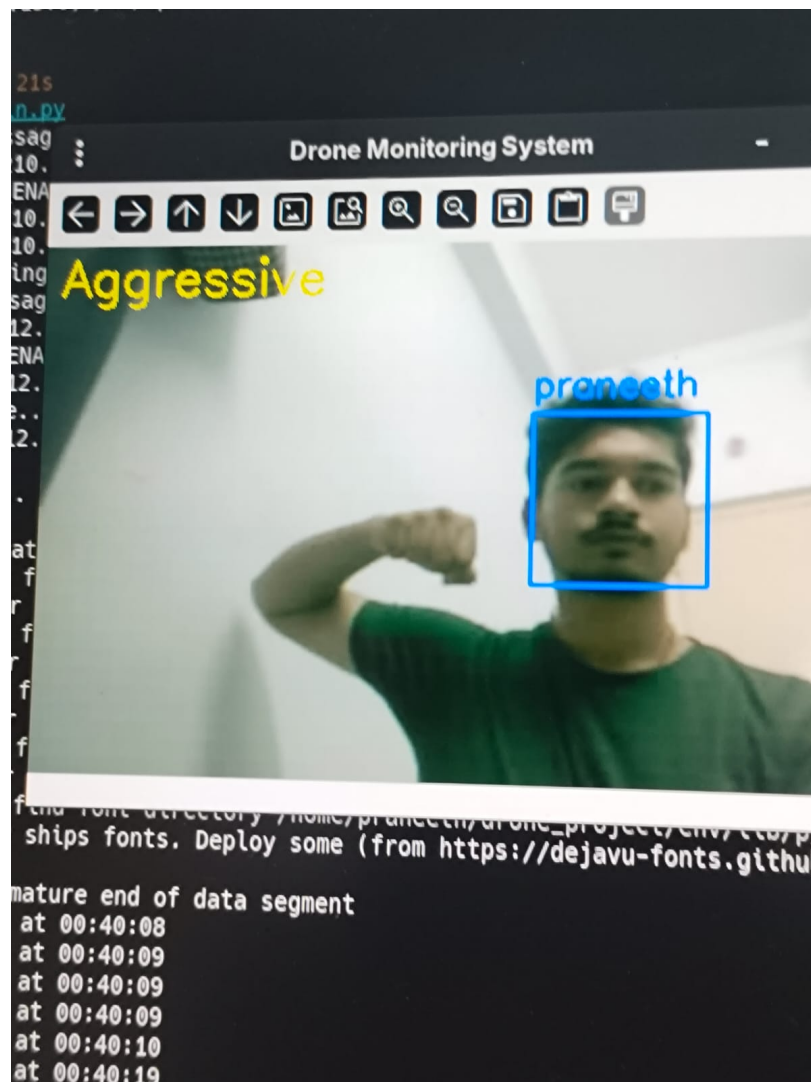


Figure 5.5: Action recognition output

When a *fight* activity is detected, the system triggers an alert, which is displayed on the output frame. This implementation provides a computationally efficient solution suitable for real-time deployment while maintaining reliable detection of significant behavioral patterns.

## 5.4 System integration

The integration of face recognition and action recognition modules enables the system to perform context-aware surveillance. Both modules operate simultaneously on incoming video frames, allowing the system to associate detected actions with identified individuals.

Initially, face recognition identifies the person present in the frame. Subsequently, the action recognition module analyzes the motion within the same frame sequence to determine the activity being performed. The outputs from both modules are combined and dis-

played together, providing a comprehensive interpretation of the scene.

For example, the system can generate outputs such as “*Vamsi – Normal*” or “*Unknown – Fight*”, which significantly enhances the usefulness of the system in surveillance scenarios.

## 5.5 Tools and technologies used

The proposed system is implemented using the following tools and technologies:

- **Programming Language:** Python
- **Libraries:**
  - OpenCV – Video acquisition and processing
  - DeepFace – Face recognition and embedding generation
  - NumPy – Numerical computations
  - Scikit-learn – Machine learning utilities
- **Development Environment:**
  - Google Colab – used for training and evaluating deep learning models (CNN-LSTM)
  - Local Python Environment – used for implementing real-time system using OpenCV, DeepFace, and action recognition modules
- **Hardware Components:**
  - Drone platform
  - ESP32-S3 camera module
  - Laptop (processing unit)

# 6

## **RESULTS AND DISCUSSIONS**

---

## 6.1 Introduction

This chapter presents the results obtained from the implementation of the proposed drone-based monitoring system. It includes the outputs of the face recognition and action recognition modules, along with performance evaluation and comparison of models.

## 6.2 Face recognition results

The face recognition module successfully detects and identifies individuals in real-time video frames. The system generates bounding boxes around detected faces and labels them with the corresponding identity.

The use of DeepFace with pretrained models enables robust identification under varying conditions such as changes in lighting and facial expressions.

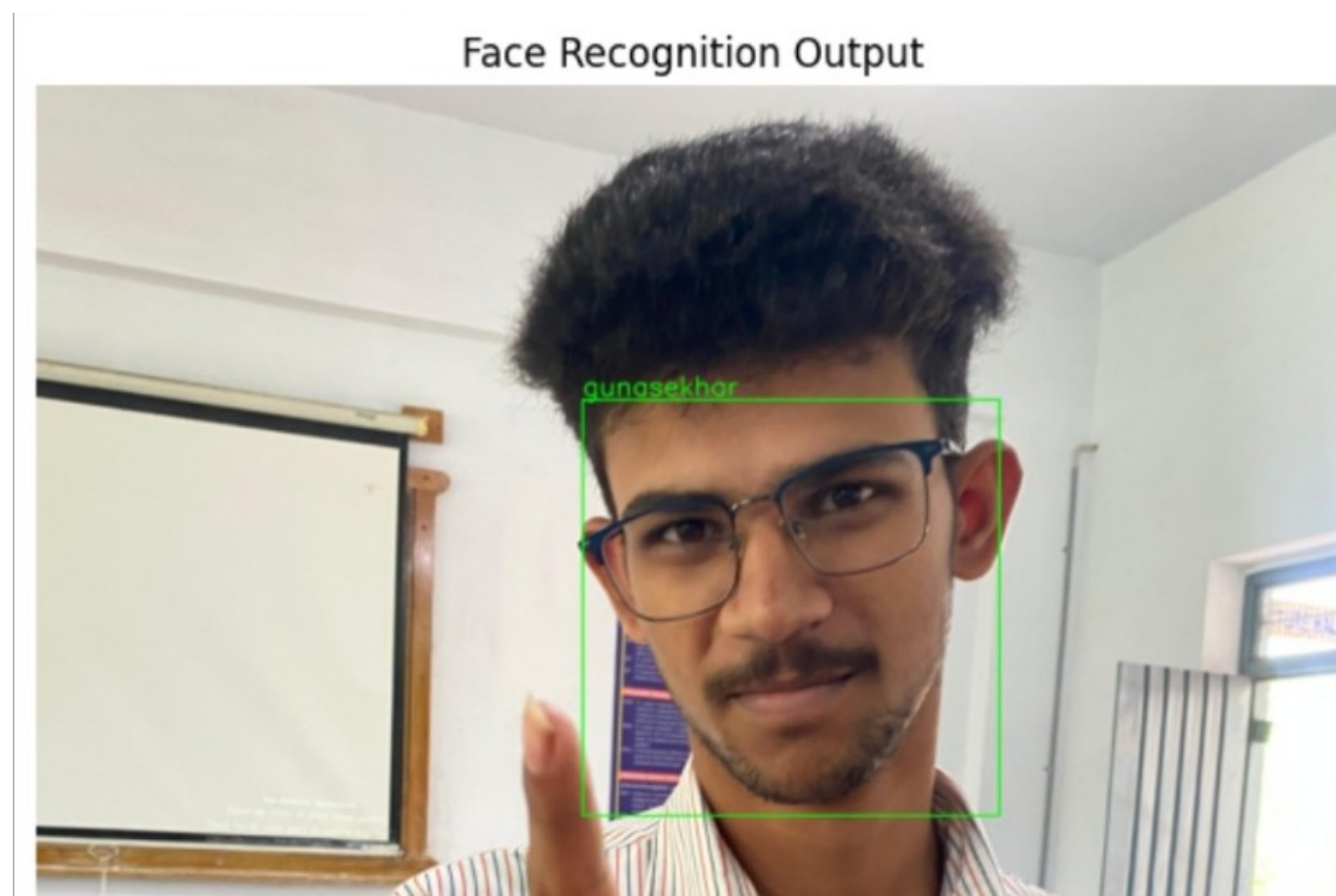


Figure 6.1: Face recognition result

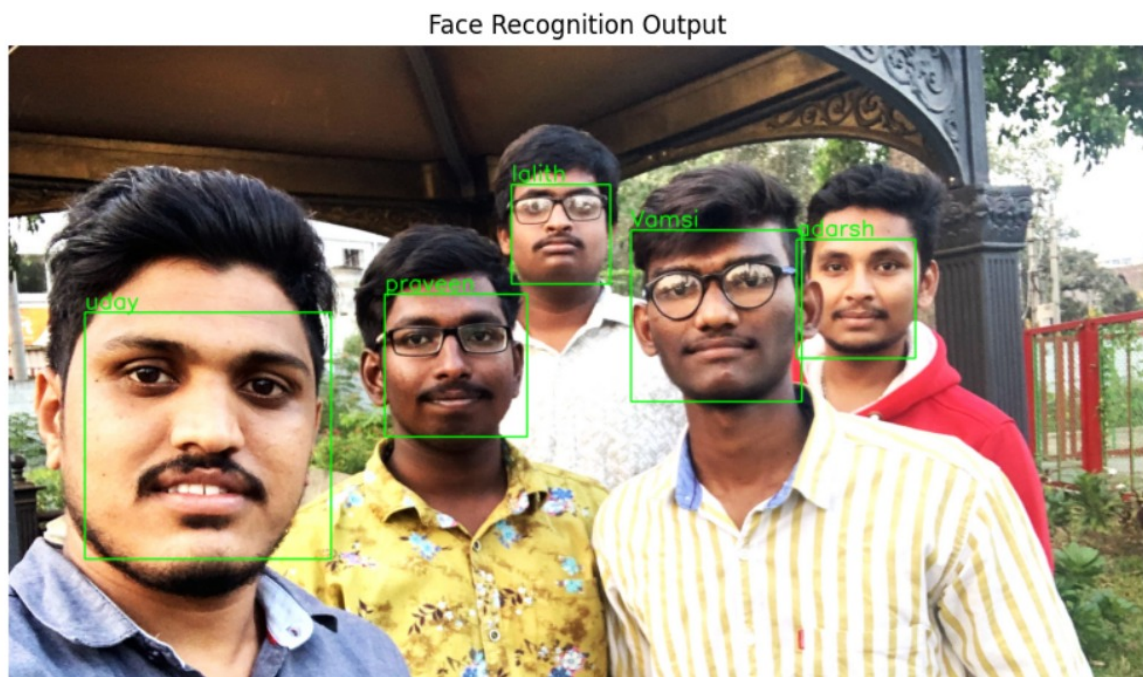


Figure 6.2: Face recognition result showing multiple individuals

**6.2.1 Face distance analysis** The distribution of distances between face embeddings for same and different individuals is analyzed to determine an appropriate threshold for recognition.

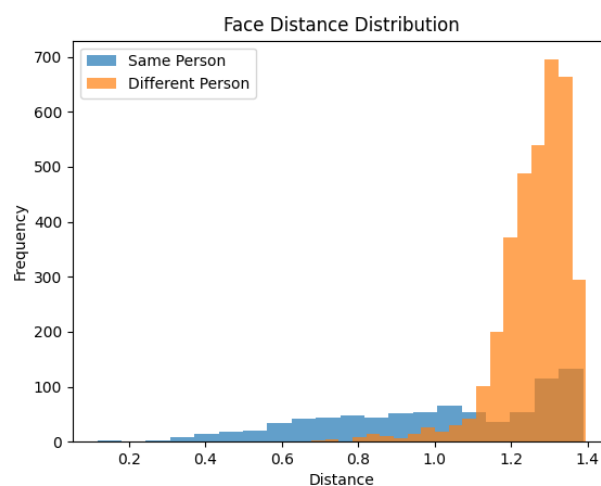


Figure 6.3: Face distance distribution for same and different individuals

**6.2.2 Embedding visualization** The face embeddings are visualized to analyze how well different individuals are separated in the feature space.

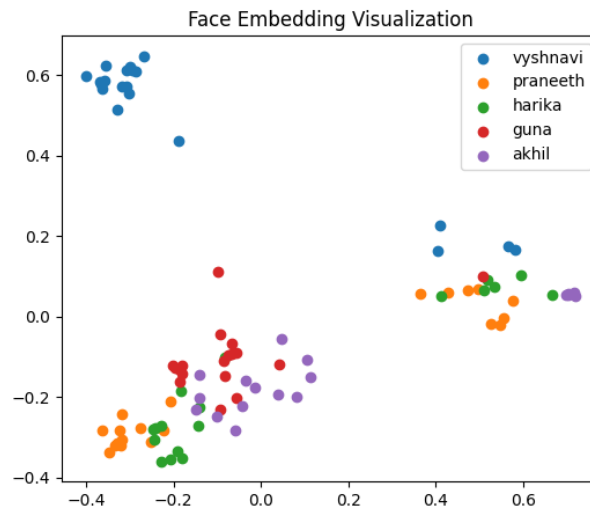


Figure 6.4: Visualization of face embeddings

### 6.3 Action recognition results

The action recognition module classifies activities into three categories: *normal*, *aggressive*, and *fight*. The system analyzes motion patterns across consecutive frames to determine the type of activity.

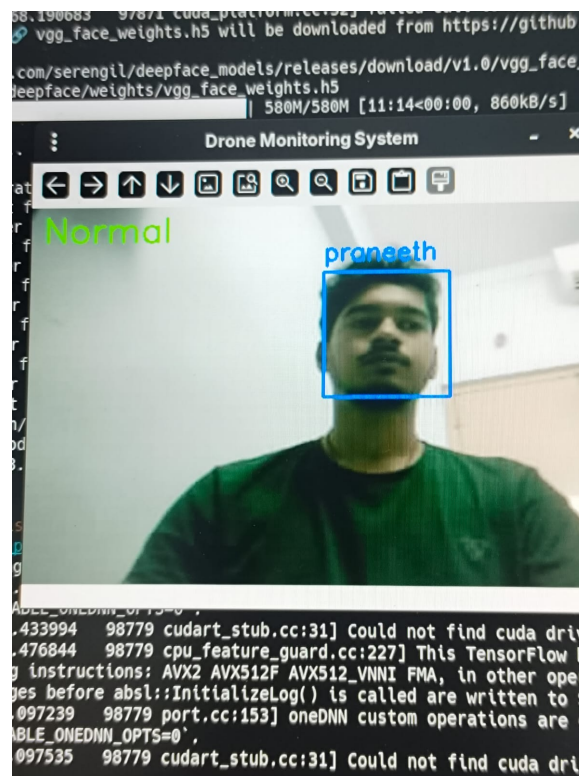


Figure 6.5: Action recognition result - A

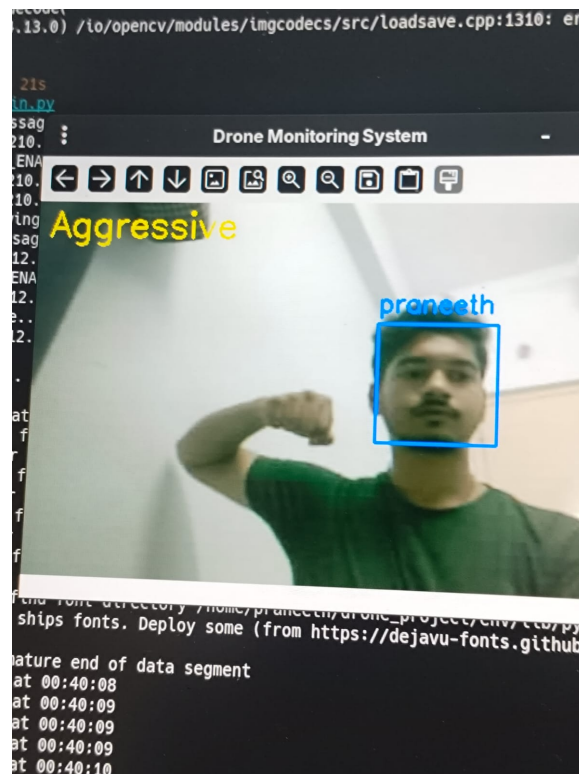


Figure 6.6: Action recognition result - B

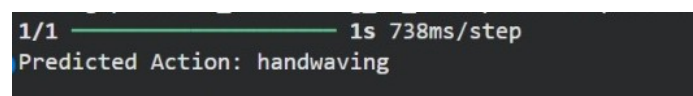


Figure 6.7: Predicted action: hand waving

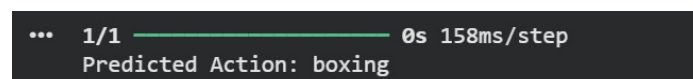


Figure 6.8: Predicted action: boxing

## 6.4 Model evaluation

The performance of the action recognition module is evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of the model's ability to correctly classify different types of activities.

Accuracy represents the overall correctness of the model by measuring the ratio of correctly predicted instances to the total number of predictions. The obtained accuracy indicates that the model performs reliably in distinguishing between different activity classes.

Precision measures the proportion of correctly predicted positive observations to the total predicted positives. A high precision value indicates that the model produces fewer false positive predictions. Recall, on the other hand, measures the proportion of actual positive cases that are correctly identified, reflecting the model's ability to detect relevant instances.

The F1-score combines both precision and recall into a single metric, providing a balanced measure of the model's performance, especially in cases where class distribution may not be uniform.

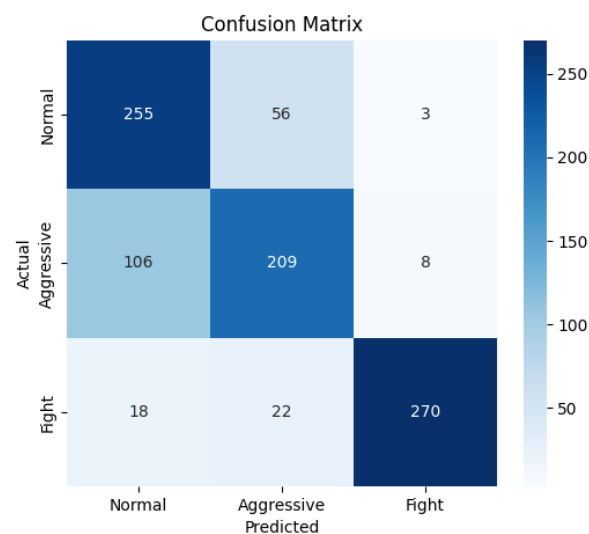


Figure 6.9: Confusion matrix for action recognition

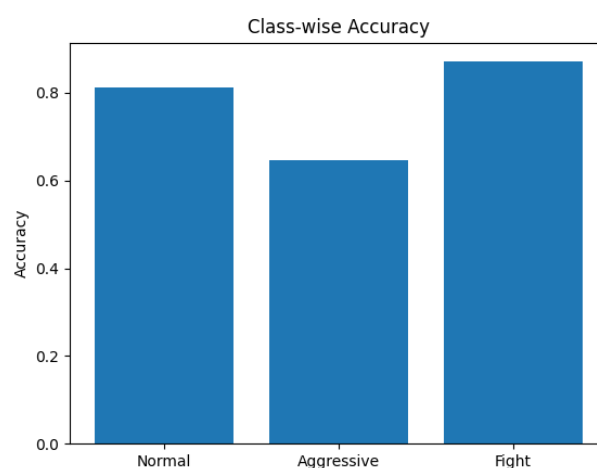


Figure 6.10: Class-wise accuracy of action recognition model

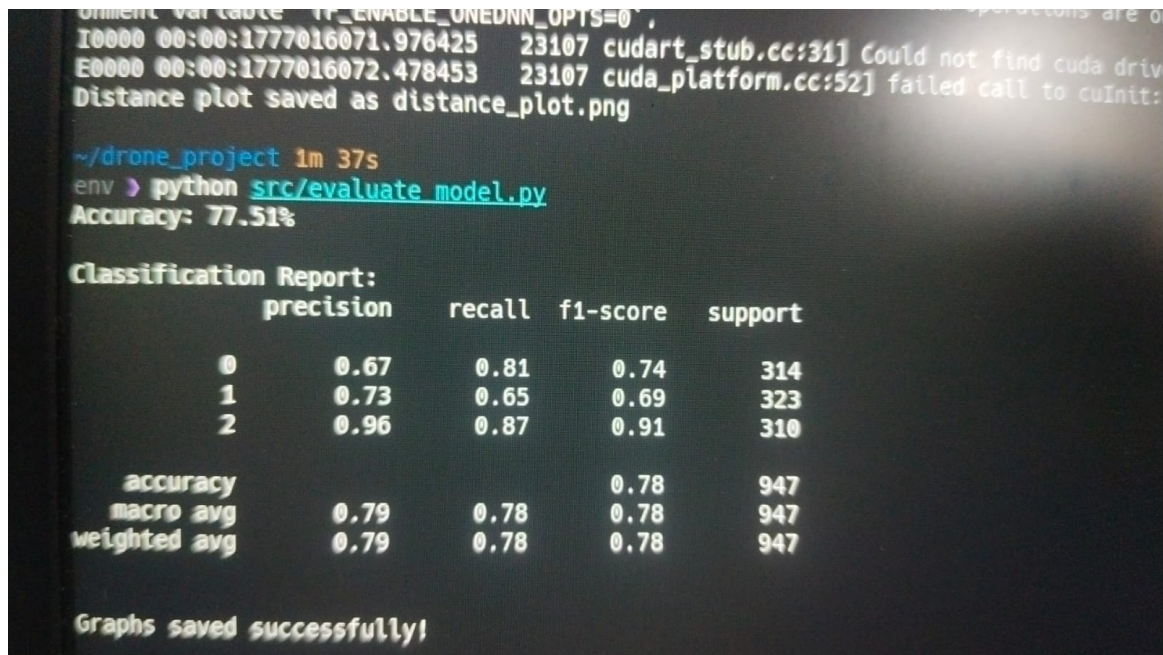


Figure 6.11: Classification report of the action recognition model

## 6.5 Deep learning model results

A deep learning-based CNN-LSTM model was evaluated for comparison with the proposed system.

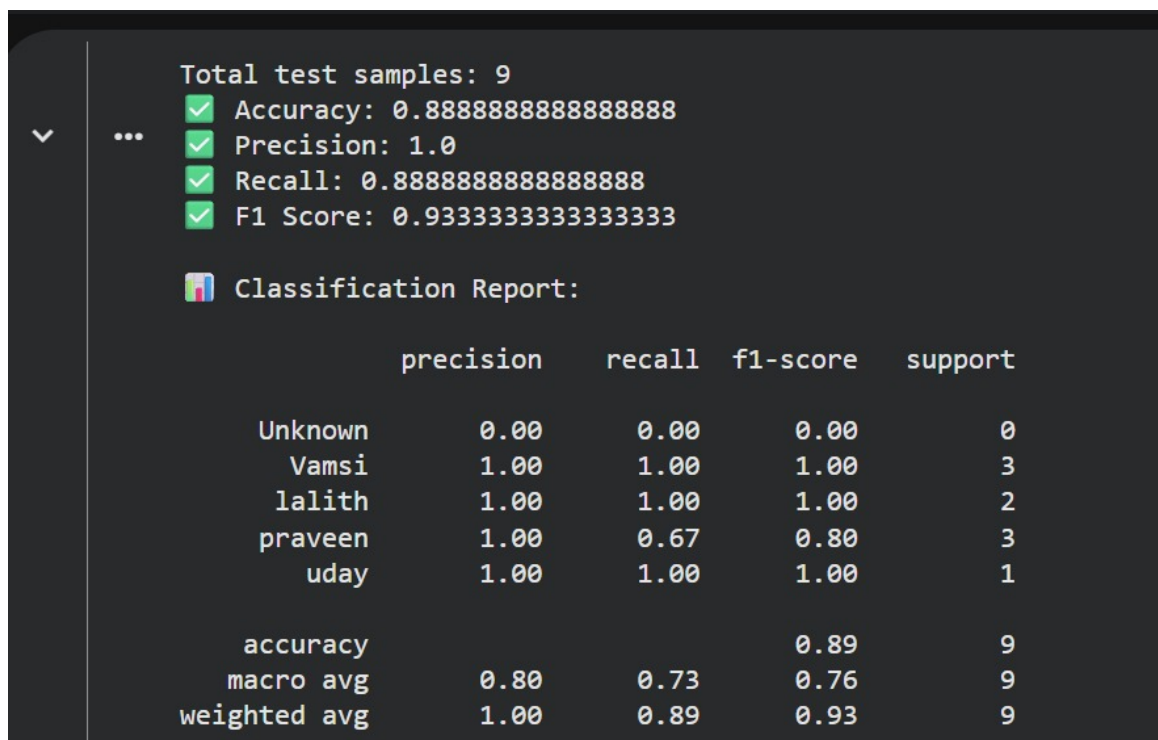


Figure 6.12: Training and validation accuracy of CNN-LSTM model

| ...          | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| handclapping | 0.80      | 0.91   | 0.85     | 22      |
| walking      | 0.71      | 0.68   | 0.70     | 22      |
| running      | 0.76      | 0.45   | 0.57     | 29      |
| boxing       | 0.94      | 0.65   | 0.77     | 26      |
| handwaving   | 0.53      | 0.80   | 0.64     | 10      |
| jogging      | 0.39      | 0.67   | 0.49     | 18      |
| accuracy     |           |        | 0.67     | 127     |
| macro avg    | 0.69      | 0.69   | 0.67     | 127     |
| weighted avg | 0.73      | 0.67   | 0.68     | 127     |

Figure 6.13: Classification report of CNN-LSTM model

## 6.6 Comparison of models

Table 6.1: Comparison of models

| Parameter             | CNN + LSTM | DeepFace + RandomForest |
|-----------------------|------------|-------------------------|
| Accuracy              | High       | Moderate to High        |
| Processing Speed      | Slow       | Fast                    |
| Computational Cost    | High       | Low                     |
| Real-time Performance | Limited    | Efficient               |
| Ease of Integration   | Difficult  | Easy                    |

## 6.7 Discussion

The results demonstrate that the proposed system is capable of performing real-time monitoring with reliable performance. The integration of face recognition and action recognition enables the system to identify individuals and analyze their behavior simultaneously.

The use of lightweight models significantly improves the system's response time and ensures smooth operation on a local computing system. Additionally, adaptive calibration and temporal smoothing enhance the stability of predictions and reduce false detections.

Although deep learning models provide higher accuracy, their computational requirements limit real-time deployment. The proposed system achieves a balance between accuracy and efficiency, making it suitable for practical applications.

## 6.8 Summary

This chapter presented the results and performance analysis of the proposed system. The system demonstrates effective real-time face recognition and action detection capabilities. A comparison with deep learning models highlights the advantages of the proposed approach in terms of efficiency and deployment feasibility.

## **CONCLUSION AND FUTURE WORK**

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## 7.1 Conclusion

This project presented the design and implementation of an AI drone-based monitoring system for campus surveillance. The system integrates real-time video acquisition with automated analysis to identify individuals and detect human activities.

The face recognition module, implemented using the DeepFace framework, enables reliable identification of individuals, while the action recognition module classifies activities based on motion patterns into categories such as *normal*, *aggressive*, and *fight*. The integration of these modules allows the system to perform context-aware monitoring by associating actions with specific individuals.

A key contribution of this work is the use of a computationally efficient approach that supports real-time processing without requiring high-end hardware. This makes the system practical for deployment in resource-constrained environments such as educational institutions.

The experimental results demonstrate that the proposed system achieves satisfactory performance in terms of detection accuracy and processing speed. Furthermore, the comparison with deep learning-based models highlights the effectiveness of the proposed approach in balancing accuracy and efficiency.

Overall, the system provides a scalable and cost-effective solution for intelligent surveillance, with potential applications in campus safety and security monitoring.

## 7.2 Future work

The proposed system can be further enhanced by incorporating additional features to improve its functionality and scalability.

- **GPS Integration:** A GPS module can be incorporated to provide the exact location of detected incidents, enabling faster response and improved situational awareness.
- **Autonomous Drone Operation:** The system can be extended to support fully autonomous navigation, reducing the need for manual control and enabling continuous monitoring.
- **Improved Accuracy:** The performance of face recognition and action recognition can be enhanced by using larger datasets, advanced models, and improved training techniques.
- **Advanced Alert System:** Integration with mobile applications or notification systems can enable real-time alerts to authorities for immediate action.

# 8

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# 9

## PUBLICATIONS

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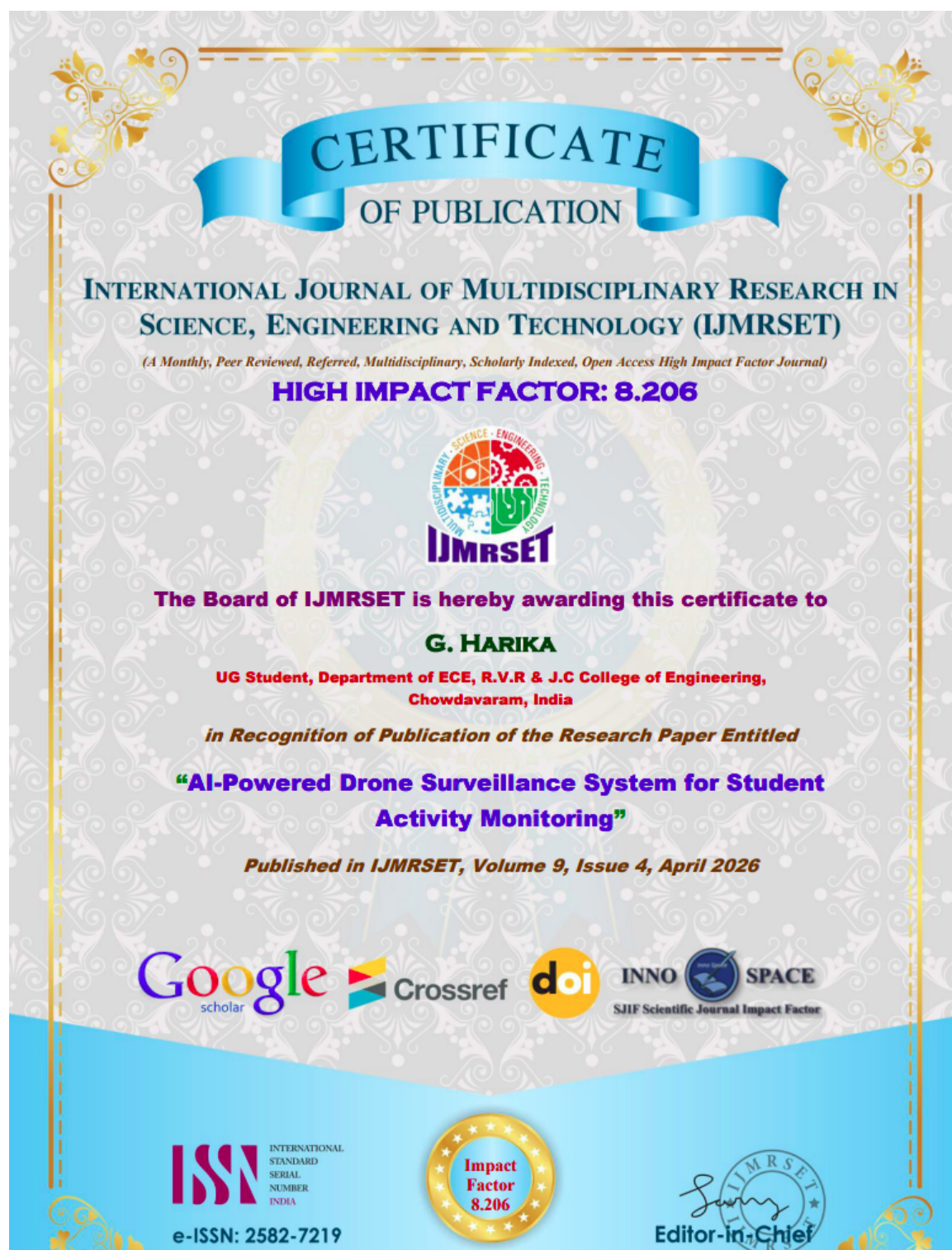
## List of Publications

- “D. Vamsi (Y22EC034), B. Praneeth Koundinya (Y22EC021), G. Harika (Y22EC047), M. Mahendra (Y21EC102)”, “AI-Powered Drone Surveillance System for Student Activity Monitoring”, International Journal of Multidisciplinary Research in Science, Engineering and Technology (IJMRSET), Volume 9, Issue 4, April 2026.

## Publication Certificates



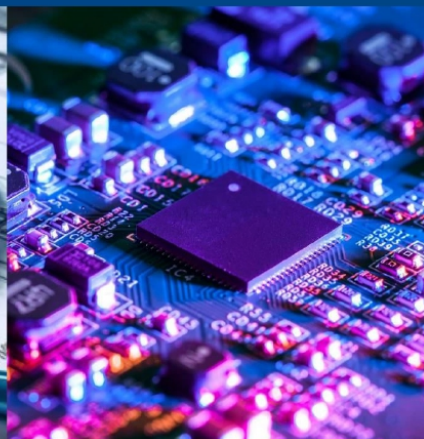






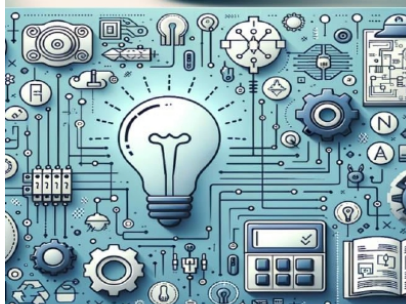


ISSN: 2582-7219



# International Journal of Multidisciplinary Research in Science, Engineering and Technology

*(A Monthly, Peer Reviewed, Refereed, Scholarly Indexed, Open Access Journal)*



Impact Factor: 8.206

Volume 9, Issue 4, April 2026

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# AI-Powered Drone Surveillance System for Student Activity Monitoring

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**ABSTRACT:** Safety and security in public areas, such as a college campus, has become a significant issue with increasing security concerns. This paper proposes an AI-based drone surveillance system which is capable of identifying humans and recognizing their actions in real-time. The system uses a drone to capture live video, which is then transmitted and processed through a ground station and cloud-based machine learning algorithms. Face recognition is done through classification using encoding-based techniques, while human activity recognition is done through a hybrid CNN and LSTM model. The system can detect activities such as boxing, running, walking, and hand movements, and sends real-time alerts through the monitoring system. The system is cost-effective and eliminates the need for human surveillance.

**KEYWORDS:** Drone Surveillance, Face Recognition, Action Recognition, CNN, LSTM, Real-Time Monitoring.

## I. INTRODUCTION

With an increased number of people and an expansion in areas like college campuses, safety and security have become a major concern. Current surveillance systems rely on fixed camera systems and human monitoring, which can be quite inefficient and time-consuming. These systems lack intelligent capabilities to understand and respond to situations, leading to delayed actions in critical situations.

Recently, Artificial Intelligence (AI) and computer vision have emerged as powerful techniques for building intelligent surveillance systems. These systems can process images and videos in real-time to identify objects, people, and human activities. However, existing intelligent surveillance systems can either perform face recognition or human activity detection. These systems often rely on fixed camera systems with limited viewing areas.

The integration of Unmanned Aerial Vehicle (UAV) technology, commonly referred to as Drones, has brought a new dimension to surveillance systems. These systems can cover large areas and offer dynamic surveillance capabilities, unlike fixed camera systems. These systems can be further enhanced with Artificial Intelligence techniques to offer efficient intelligent surveillance systems.

This paper proposes a surveillance system based on Artificial Intelligence techniques and Drones. The proposed system integrates face recognition and human activity detection for efficient decision-making capabilities. The proposed system can be quite efficient in offering intelligent surveillance capabilities without relying on human supervision. The proposed system utilizes a drone for capturing video streams, which are further processed at a ground station and in the cloud using machine learning techniques for face and human activity detection. The proposed system can be quite efficient in offering intelligent surveillance capabilities without relying on human supervision.



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### II. LITERATURE REVIEW

Here we propose an AI-based drone surveillance system, which is capable of real-time human identification and activity recognition. The proposed system would utilize computer vision, deep learning, and drones, making it an intelligent surveillance system. Various research works have been carried out on AI-based surveillance, focusing more on improving accuracy.

Face recognition systems have been widely utilized in various applications, including human identification. Face recognition systems use encoding techniques to identify individuals.

Some of the advantages of face recognition systems are:

- Highly accurate in identifying human faces
- Fast, as encoding techniques are employed
- Can be utilized in real-time applications

Some of the disadvantages of face recognition systems are:

- Not capable of recognizing human activities or behavior
- Accuracy may be reduced depending on light conditions

Human activity recognition systems have been proposed using deep learning techniques, including Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM). These systems utilize video analysis to identify human activities, including walking, running, or fighting.

The advantages of the activity recognition system are as follows:

- Multiple human actions can be classified
- Efficient use of temporal and spatial features
- Suitable for behavior monitoring

However, the system has some challenges as follows:

- Does not give the identity of the individuals
- Requires a large dataset
- Computational intensive

Surveillance systems using drones are important due to the ability of the system to offer a dynamic and wide area of surveillance. This is because the system can move freely and capture video images in real time from different angles.

However, the advantages of the drone-based surveillance system are as follows:

- Wide area of surveillance
- Can capture video images in real time
- Does not depend on infrastructure

However, the system has some challenges as follows:

- Does not integrate with intelligent analysis
- Depends on other systems

With the recent advancements in AI-based surveillance systems, the integration of multiple technologies can offer a more efficient system. This can be achieved by integrating face recognition and activity recognition with drone-based surveillance systems to offer a more efficient system. The proposed system integrates multiple technologies to offer a more efficient system in the real world. This system can efficiently identify humans and monitor behavior in real time. This can improve the efficiency of the system and the decision-making ability of the system. This can reduce the need for supervision by humans.



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### III. PROPOSED METHOD

#### Problem Statement:

The traditional surveillance system has been seen to be less efficient, especially due to its limitation to the coverage area and its dependency on human intervention. The traditional system lacks intelligence and hence cannot perform an analysis on the current situations, especially when identifying individuals and their actions.

In addition, most of the existing intelligent surveillance systems using AI are found to be focused only on the recognition of faces or the actions performed, and this is found to be incomplete. The limitation is the major factor that reduces the efficiency of the system in handling the real-time situations.

#### Objectives

The objectives of the proposed system are as follows:

- Design and develop an intelligent system using drones for video surveillance
- Design and develop an intelligent system using face recognition and encoding
- Design and develop an intelligent system using action recognition and CNN/LSTM
- Real-time alerts at the monitoring station
- Cost-effective and efficient surveillance system

#### System Architecture Overview

The proposed system includes three main components:

1. Drone Unit:  
It records real-time video feed and covers dynamic areas.
2. Base Station:  
It receives video data from the drone and acts as an interface for monitoring.
3. Cloud-Based Processing:  
It executes machine learning models for face and action detection.

#### Working Methodology

The working methodology of the proposed system is explained below:

1. The drone records video feed in real-time from the surveillance area.
2. It sends video data to the base station.
3. It processes video frames and sends them for face and action detection.
4. Face detection recognizes individuals based on stored encodings.
5. Action detection recognizes human actions using a CNN-LSTM model.
6. It generates an alert message based on detection outcomes.
7. It continuously executes this process for real-time video monitoring.

#### Architecture Diagram

1. **System Initialization:** The system is configured by powering the drone and linking it to the ground station. The required modules are now ready for processing.



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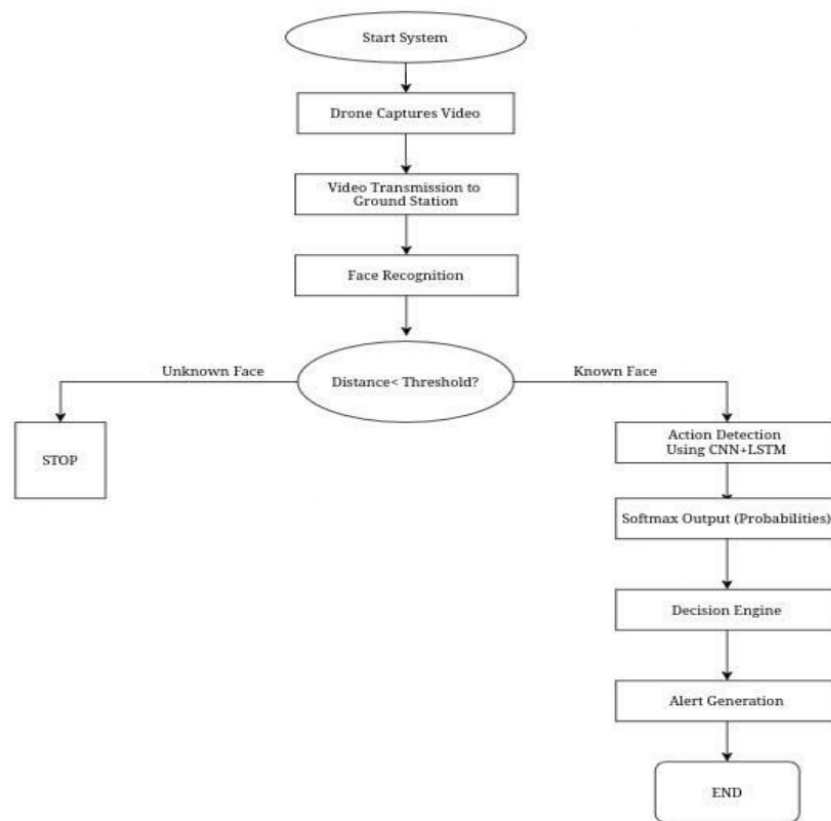


Fig.1. Architecture diagram

2. **Video Capture:** The drone captures continuous live video feed from the surveillance region using the camera installed on it.
3. **Video Transmission:** The video feed is now transmitted to the ground station (laptop) via ESP32 S3 module.
4. **Frame Processing:** The system converts the received video feed into individual frames then the frames are resized and normalized for processing.
5. **Face Recognition:** The faces are recognized within individual frames and then compared for identity recognition.
6. **Decision Making :** The recognized face is then checked for matching identity based on a threshold value. If there is no match, then continuous processing occurs.
7. **Action Recognition:** If a match is found for a recognized face, then human activity recognition occurs based on individual frames using a CNN-LSTM model.
8. **Activity Classification:** The activity is then recognized based on individual probabilities for different actions. The action for which the probability is the highest is selected as the recognized activity.
9. **Decision Engine:** The recognized identity and activity are then used to make a decision based on individual information.
10. **Alert Generation:** A real-time alert message is displayed on the display panel for the activity detected within the region.
11. **Continuous Monitoring:** The above steps are repeated for continuous processing.



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### IV. IMPLEMENTATION OF THE SYSTEM

#### Requirements for Software and Hardware

##### Software Requirements:

##### Python and Google Colab:

The proposed system has been implemented using Python as a programming language. The system runs on Google Colab. Google Colab has enough computational capabilities to train and execute the machine learning models. Google Colab also facilitates easy handling of datasets, execution of models, and real-time inference.

##### OpenCV:

The system uses OpenCV for video processing. OpenCV facilitates capturing images from videos, resizing images, etc. OpenCV ensures that images are properly processed for execution through the machine learning models.

##### Face Recognition Library:

The face recognition system is performed using a library called \*face\_recognition\*, which uses a pre-trained deep learning model. The system uses a technique called Histogram of Oriented Gradients (HOG) for detecting faces. The system extracts features from faces as 128D vectors. The system stores these vectors for execution. During execution, it uses Euclidean distance to compare vectors for identifying faces.

##### TensorFlow and Keras:

The system uses TensorFlow and Keras for implementing the action recognition system. The system uses a hybrid model that integrates CNN with LSTM for recognizing human activities. CNN extracts spatial features from images, while LSTM extracts temporal features from a sequence of images.

##### Face Recognition Implementation:

The dataset for known people is created using multiple images. The images are processed to get facial encodings, which are stored for future use. During the execution phase, the faces in each video frame are identified, and encodings are obtained. The encodings obtained from the video frames are compared with the stored encodings using the distance method. If the distance is less than a threshold, the individual is identified; else, the face is marked as unknown.

##### Action Recognition Implementation:

The action recognition model is implemented using a sequence-based deep learning approach. The video data is divided into frames. Each frame is resized and normalized. A specified number of frames are used for the model.

The CNN layers identify the spatial features for each frame. The LSTM layer recognizes the sequence of frames to identify the action. The output is acquired using a softmax layer. Output of Action Recognition is the probabilities for each action: boxing, running, walking, hand movement, etc. The action with the maximum probability is selected as the action being performed.

##### System Integration:

The face recognition as well as action recognition modules are integrated with a unified system. The video feed from the drone is processed at the ground station, where face recognition occurs first. If a known person is recognized, then action recognition follows. According to the recognized action, a real-time warning message pops up on the monitoring screen.

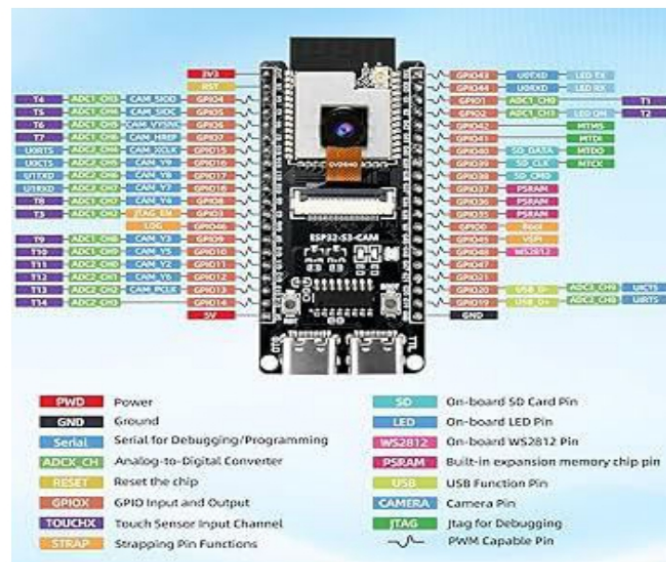
#### Hardware Requirements



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### 1. ESP32-S3 Module:



The primary hardware component in this system is ESP32 S3 Module. It is used for receiving and transmitting video data from the camera module to the ground station. The module is also used for Wi-Fi connection, which is necessary for real-time communication with the drone.

#### 1. Drone Platform:

The drone is used as a mobile surveillance component in this system. The drone is used for carrying the ESP32-S3 module with the camera module for real-time video transmission from different angles. The drone is useful for covering more area with respect to static systems.

#### 2. In-built Camera Module:

The camera module is used for receiving live video feed from different sources and is used for providing inputs to the machine learning models for action and face detection.

#### 3. Ground Station (Laptop):

The laptop is used as a receiving component in this system, which is connected to the drone using Wi-Fi for receiving live video feed from the drone and executing face detection and action detection models.

## V. RESULT

To test the efficacy of the proposed drone surveillance system based on artificial intelligence (AI), several tests were carried out with video footage. The AI-based surveillance system showed positive results in human recognition and classification of human activities.

In order to test the performance of the face recognition module of the proposed drone surveillance system, several tests were carried out with faces of multiple individuals. The system successfully detected and identified people by encoding faces. The output result of the test is shown in Fig. 3, in which multiple faces have been detected successfully by the system.



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Face Recognition Output

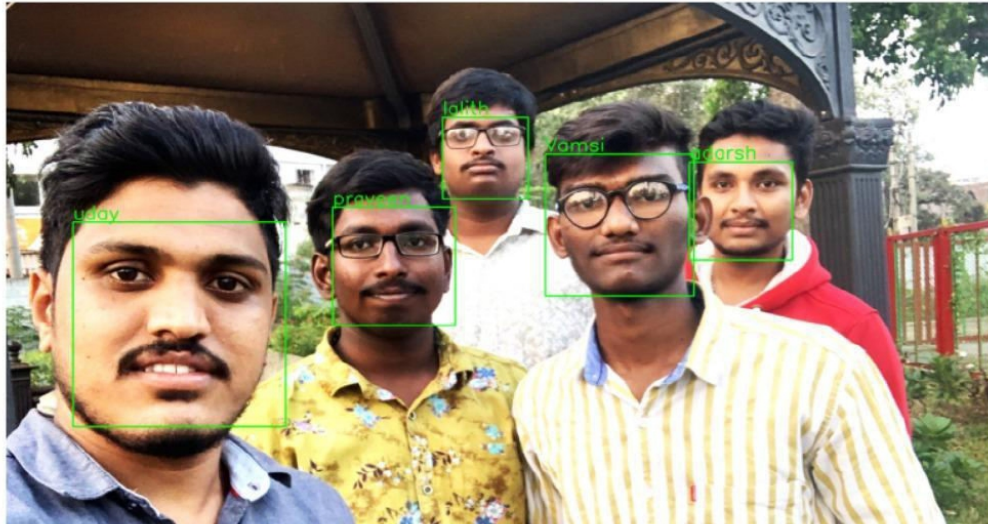


Fig. 3: Output Result from the Face Recognition Module (Multiple Individuals)

Additionally, the face recognition module was also tested with input from a single person. The system was able to recognize the single person with high efficiency, as illustrated in Fig. 4.

Face Recognition Output

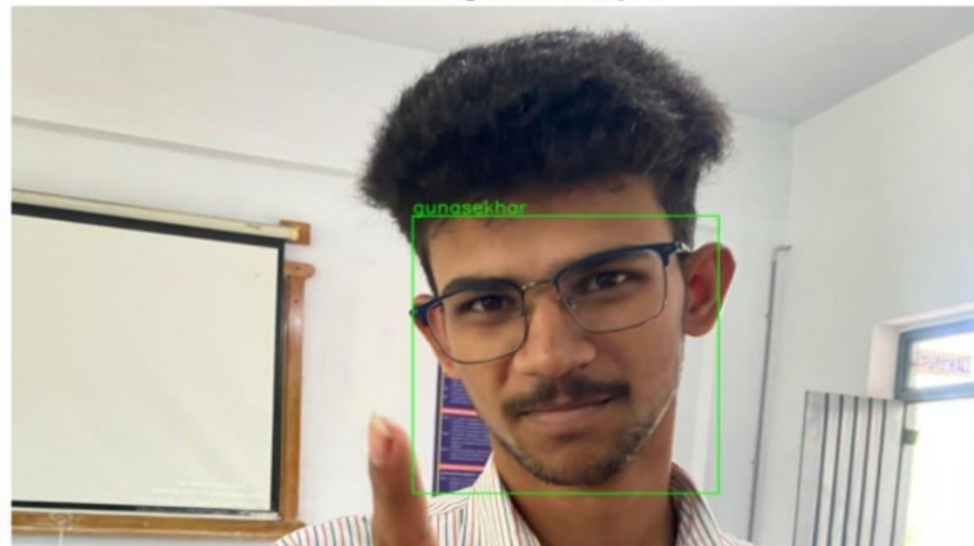


Fig. 4: Output Result from Face Recognition Module (Single Person Detection)



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### Action Recognition

The performance of the action recognition module has been tested with action samples. The CNN-LSTM model classified human actions such as boxing, handwaving, running, and walking successfully, as presented in Fig. 5.

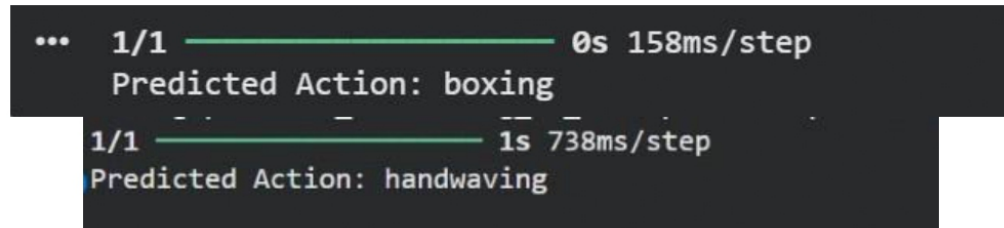
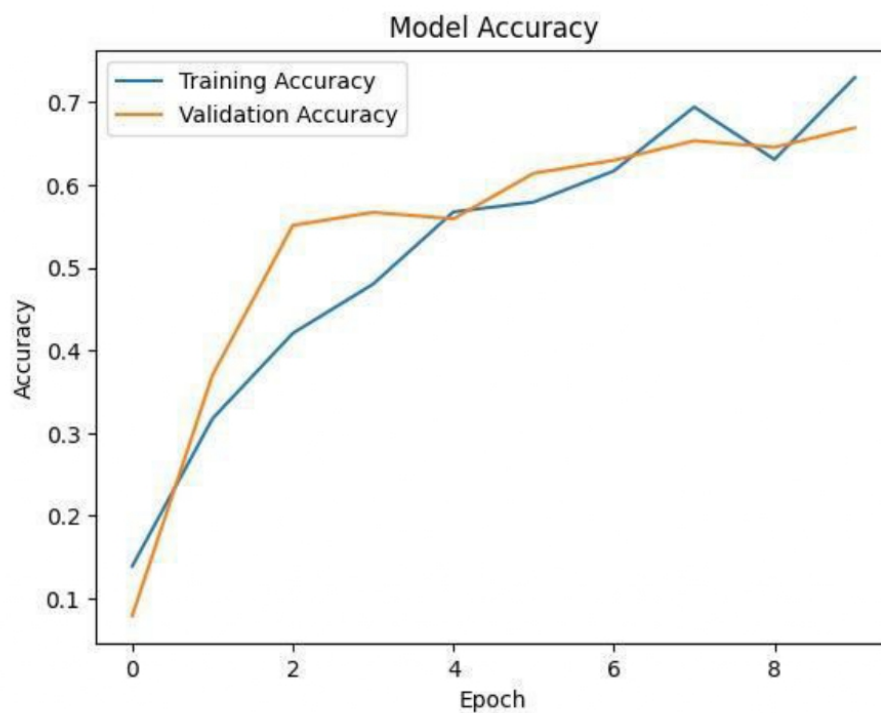


Fig. 5: Output of Action Recognition System for Identified Activity

The training performance of the action recognition system was evaluated based on the performance metrics of accuracy. The accuracy of the model was increasing in both the training and validation sets with respect to various epochs, which is depicted in Fig. 6 below.





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| ...          | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| handclapping | 0.80      | 0.91   | 0.85     | 22      |
| walking      | 0.71      | 0.68   | 0.70     | 22      |
| running      | 0.76      | 0.45   | 0.57     | 29      |
| boxing       | 0.94      | 0.65   | 0.77     | 26      |
| handwaving   | 0.53      | 0.80   | 0.64     | 10      |
| jogging      | 0.39      | 0.67   | 0.49     | 18      |
| accuracy     |           |        | 0.67     | 127     |
| macro avg    | 0.69      | 0.69   | 0.67     | 127     |
| weighted avg | 0.73      | 0.67   | 0.68     | 127     |

Fig. 6: Training Performance of the Action Recognition System with Respect to Accuracy

The system performed well under live situations. Combining face recognition and action recognition resulted in intelligent monitoring and hence could be used in areas such as campus safety surveillance.

## VI. CONCLUSION

This paper demonstrates an AI-based drone surveillance system that is capable of real-time identification of humans and recognition of their activities. The proposed system has improved situational awareness, as opposed to traditional means of surveillance. The use of a drone for surveillance is advantageous, as opposed to camera-based systems. The face recognition component of the proposed system is effective in identifying individuals through encoding-based approaches. Similarly, the action recognition component of the proposed system is effective in recognizing activities of humans through a CNN-LSTM-based approach.

The proposed system is cost-effective, efficient, and has potential applications in campus security, public security, and other areas. In conclusion, the proposed system has demonstrated the potential for artificial intelligence to be used in conjunction with drones for intelligent and automated systems.

## VII. RECOMMENDATIONS FOR FUTURE WORK:

The proposed system could be improved to function efficiently in different environments. Future work could be directed toward improving the accuracy of the system. In addition, future work could be directed toward implementing real-time processing on edge devices, as opposed to cloud-based systems and automating the operation of Drone surveillance.

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© 2026 IJMRSET | Volume 9, Issue 4, April 2026|

DOI:10.15680/IJMRSET.2026.0904131

ISSN: 2582-7219

| www.ijmrset.com | Impact Factor: 8.206| ESTD Year: 2018|



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